AI-Powered High-Volatility Algorithmic Trading System

Complete Blueprint — Interactive Brokers + Options + Futures + 0DTE

Version 2.0 — Full Rewrite with IBKR, Options Greeks, Futures, and Aggressive Strategies

---

TABLE OF CONTENTS

1. System Overview & Architecture
2. Technology Stack & Cost Breakdown
3. Hardware & Software Setup (Step-by-Step)
4. Module 1: Interactive Brokers Integration
5. Module 2: Data Pipeline (Equities + Options + Futures)
6. Module 3: Feature Engineering (Extended for Options/Futures)
7. Module 4: AI/ML Brain (Ollama + Local Models)
8. Module 5: Trading Strategies (12 Strategies Including 0DTE)
9. Module 6: Options-Specific Intelligence
10. Module 7: Risk Management Engine (Options-Aware)
11. Module 8: Execution Engine (IBKR-Specific)
12. Module 9: Monitoring & Adaptive Layer
13. Module 10: Backtesting Framework
14. Module 11: System Configuration & Project Structure
15. Module 12: Safety Rules & Operational Procedures
16. Implementation Priority Order (Step-by-Step Build Plan)

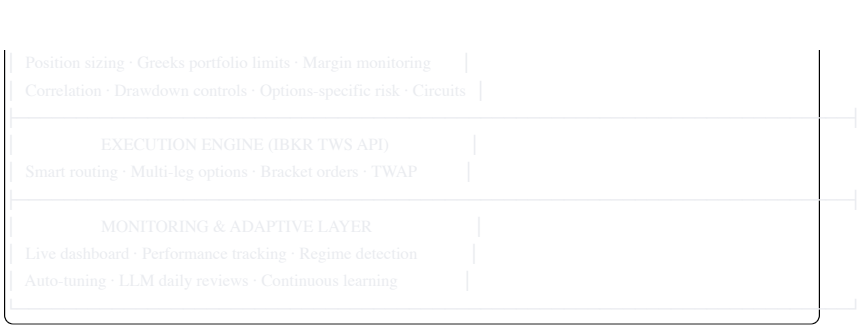
---

1. SYSTEM OVERVIEW & ARCHITECTURE

This system is a fully autonomous AI-enhanced algorithmic trading platform that trades equities, options (including 0DTE), and futures through Interactive Brokers. It combines traditional quantitative algorithms with locally-run AI models (via Ollama) for market analysis, trade decisions, and continuous learning. The entire stack is free to operate.

Master Architecture





---

2. TECHNOLOGY STACK & COST BREAKDOWN

Complete Stack

Component	Technology	Cost
Language	Python 3.11+	Free
Brokerage	Interactive Brokers (paper account)	Free
Market Data	IBKR paper data + yfinance + FRED	Free
Options Data	IBKR options chains (real-time in paper)	Free
Futures Data	IBKR futures data (paper)	Free
AI/LLM	Ollama (Llama 3.1 8B, Mistral 7B, Phi-3)	Free (local)
ML Framework	PyTorch + scikit-learn + XGBoost + LightGBM	Free
NLP/Sentiment	FinBERT (Hugging Face, local)	Free
Options Pricing	QuantLib-Python + py_vollib	Free
Technical Analysis	TA-Lib + pandas-ta	Free
IBKR Python Client	ib_insync (primary) + ibapi (backup)	Free
Data Storage	SQLite + Parquet	Free
Scheduling	APScheduler	Free
Visualization	Plotly + Dash	Free
News	IBKR news + RSS feeds	Free
Backtesting	Custom engine + vectorbt	Free
Options Analytics	mibian + py_vollib	Free

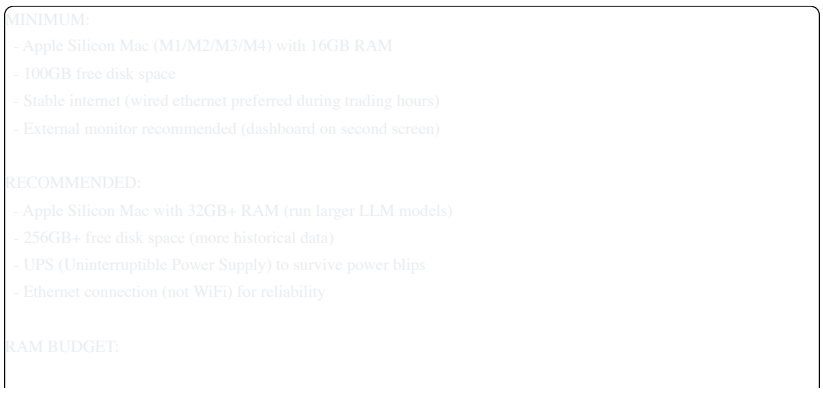
Total Monthly Cost: $0

The only costs are electricity (~$5-10/month running your Mac) and internet. No paid API subscriptions are required.

---

3. HARDWARE & SOFTWARE SETUP (STEP-BY-STEP)

3.1 Hardware Requirements



- Ollama (Llama 3.1 8B): ~5-6GB
- Python trading system: ~2-4GB
- IBKR TWS/Gateway: ~1-2GB
- FinBERT model: ~1GB
- OS + other: ~3-4GB
- Total: ~13-17GB (16GB works, 32GB is comfortable)

3.2 Software Installation (Step-by-Step for macOS)

Follow these steps in exact order:

```bash
```

```
# ============================================================
# STEP 1: Install Homebrew (macOS package manager)
# ============================================================
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

# Add to PATH (Apple Silicon Macs)
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv)"

# ============================================================
# STEP 2: Install Python 3.11
# ============================================================
brew install python@3.11
# Verify:
python3.11 --version

# ============================================================
# STEP 3: Install system dependencies
# ============================================================
brew install ta-lib       # Technical analysis C library
brew install cmake        # Needed for some ML packages
brew install wget         # File downloads

# ============================================================
# STEP 4: Install Ollama (Local AI)
# ============================================================
brew install ollama

# Start Ollama service (runs in background)
ollama serve &

# Pull AI models (this downloads several GB — be patient)
ollama pull llama3.1:8b          # Primary reasoning (4.7GB)
ollama pull mistral:7b           # Secondary/ensemble (4.1GB)
ollama pull nomic-embed-text     # Embeddings (274MB)

# Verify Ollama is working:
ollama run llama3.1:8b "Say hello in one sentence"
# You should get a response. Press Ctrl+D to exit.

# ============================================================
# STEP 5: Install Interactive Brokers TWS or IB Gateway
# ============================================================
# Option A: TWS (Trader Workstation) — Full GUI, good for monitoring
#   Download from: https://www.interactivebrokers.com/en/trading/tws.php
#   Install the "Latest" version for macOS
#
# Option B: IB Gateway — Lightweight, no GUI, better for pure algo trading
#   Download from: https://www.interactivebrokers.com/en/trading/ibgateway-stable.php
#   Install for macOS
#
# RECOMMENDATION: Start with TWS (you can see what's happening).
#                 Switch to IB Gateway later for production.
#
# After installing, launch TWS/Gateway and:
#   1. Log in with your IBKR paper trading credentials
#   2. Go to File → Global Configuration → API → Settings
#   3. Check "Enable ActiveX and Socket Clients"
#   4. Set Socket Port to: 7497 (paper) or 7496 (live — DON'T use live yet)
#   5. Check "Allow connections from localhost only"
#   6. Uncheck "Read-Only API" (we need to place orders)
#   7. Click Apply and OK

# ============================================================
# STEP 6: Create IBKR Paper Trading Account
# ============================================================
# If you don't have an IBKR account yet:
#   1. Go to https://www.interactivebrokers.com
#   2. Click "Open Account"
#   3. You can open a paper trading account for free
#   4. Or if you have a real account, enable paper trading in
#      Account Management → Settings → Paper Trading Account
#
```

```
# Paper account gives you $1,000,000 in fake money
# Full access to: stocks, options, futures, forex

# ============================================================
# STEP 7: Create project directory and virtual environment
# ============================================================
mkdir -p ~/ai-trading-system
cd ~/ai-trading-system

python3.11 -m venv venv
source venv/bin/activate

# Verify you're in the virtual environment:
which python
# Should show: /Users/yourname/ai-trading-system/venv/bin/python

# ============================================================
# STEP 8: Install Python packages
# ============================================================

# Create requirements.txt (copy the contents from Section 11.3)
# Then:
pip install --upgrade pip
pip install -r requirements.txt

# Install TA-Lib Python wrapper (requires brew ta-lib from Step 3)
pip install TA-Lib

# ============================================================
# STEP 9: Create project structure
# ============================================================
mkdir -p config src/data src/features src/ai src/ml src/strategies
mkdir -p src/options src/risk src/execution src/monitoring src/utils
mkdir -p models data/parquet logs notebooks tests

# Create __init__.py files
find src -type d -exec touch {}/__init__.py \;

# ============================================================
# STEP 10: Create environment file
# ============================================================
cat > config/.env << 'EOF'
# Interactive Brokers
IBKR_HOST=127.0.0.1
IBKR_PORT=7497
IBKR_CLIENT_ID=1
IBKR_ACCOUNT=DU1234567

# FRED (get free key: https://fred.stlouisfed.org/docs/api/api_key.html)
FRED_API_KEY=your_fred_api_key_here

# Ollama
OLLAMA_BASE_URL=http://localhost:11434

# Reddit (optional, get free: https://www.reddit.com/prefs/apps)
REDDIT_CLIENT_ID=
REDDIT_CLIENT_SECRET=
REDDIT_USER_AGENT=trading-bot/1.0
EOF

echo "config/.env" >> .gitignore
echo "models/" >> .gitignore
echo "data/" >> .gitignore
echo "logs/" >> .gitignore
echo "__pycache__/" >> .gitignore
echo "*.pyc" >> .gitignore

# ============================================================
# STEP 11: Verify everything works
# ============================================================

# Test IBKR connection (TWS/Gateway must be running):
python -c "
```

```
from ib_insync import *
ib = IB()
try:
    ib.connect('127.0.0.1', 7497, clientId=1)
    print(f'Connected! Account: {ib.managedAccounts()}')
    print(f'Net Liquidation: {ib.accountSummary()[0].value if ib.accountSummary() else \"N/A\"}')
    ib.disconnect()
    print('IBKR connection: OK')
except Exception as e:
    print(f'IBKR connection failed: {e}')
    print('Make sure TWS/Gateway is running with API enabled on port 7497')
"

# Test Ollama:
python -c "
import requests
r = requests.post('http://localhost:11434/api/generate',
    json={'model': 'llama3.1:8b', 'prompt': 'Say OK', 'stream': False})
print(f'Ollama response: {r.json()[\"response\"][:50]}')
print('Ollama: OK')
"

# Test TA-Lib:
python -c "
import talib
print(f'TA-Lib version: {talib.__ta_version__}')
print('TA-Lib: OK')
"

# Test PyTorch:
python -c "
import torch
print(f'PyTorch: {torch.__version__}')
print(f'MPS (Apple GPU) available: {torch.backends.mps.is_available()}')
print('PyTorch: OK')
"

print("Setup complete! All systems verified.")
```

3.3 IBKR Paper Account Configuration

```
After logging into TWS paper trading, configure these settings:

1. API Settings (File → Global Configuration → API → Settings):
   - Enable ActiveX and Socket Clients: ✓
   - Socket port: 7497
   - Allow connections from localhost only: ✓
   - Read-Only API: ✗ (unchecked — we need write access)
   - Master API client ID: Leave blank
   - Precautions → Bypass Order Precautions for API Orders: ✓

2. Market Data Subscriptions (paper account):
   - Paper accounts get delayed data by default (15 min)
   - For real-time paper data, you need to subscribe to market data
     in your REAL account (even if you only paper trade)
   - Minimum recommended subscriptions for this system:
     • US Securities Snapshot and Futures Value Bundle ($10/mo)
     • US Equity and Options Add-On Streaming Bundle ($4.95/mo)
   - These are the ONLY potential costs. You can start without them
     using delayed data, but real-time is much better for day trading.
   - NOTE: Market data fees are waived if you generate $30+/month
     in commissions on a real account.

3. Trading Permissions (Account Management → Settings):
   - Stocks: ✓
   - Options: ✓ (requires options trading approval)
   - Futures: ✓ (requires futures trading approval)
   - United States: ✓

4. Paper Account Capital:
   - Default: $1,000,000
   - You can request adjustment through Account Management
   - For this system, $100,000 is a realistic starting point
     (reset your paper account to $100,000 for realistic testing)
```

4. MODULE 1: INTERACTIVE BROKERS INTEGRATION

4.1 IBKR Connection Manager

```
ibrary: ib_insync (preferred) — async-friendly Python wrapper for IBKR TWS API

Why ib_insync over raw ibapi:
- Much cleaner async/await API
- Automatic reconnection
- Pythonic objects instead of callback spaghetti
- Built-in event system
- Maintained and well-documented

nstall: pip install ib_insync

onnection Architecture:
- ONE persistent connection to TWS/Gateway
- Connection must stay alive during market hours
- ib_insync handles heartbeats automatically
- Implement reconnection logic for drops:
  - Retry every 5 seconds, up to 10 attempts
  - If reconnection fails, alert + close all positions
  - Log every disconnect/reconnect event

lass: IBKRClient

ore Methods:
connect() → Establish connection to TWS/Gateway
disconnect() → Graceful disconnect
reconnect() → Attempt reconnection with backoff
is_connected() → Check connection health

# Account
get_account_summary() → Net liquidation, buying power, margin, etc.
get_positions() → All current positions with P&L
get_portfolio() → Full portfolio with Greeks (for options)

# Market Data
get_realtime_bars(contract, bar_size) → Streaming real-time bars
get_historical_bars(contract, duration, bar_size) → Historical OHLCV
get_market_data(contract) → Snapshot: bid, ask, last, volume
get_options_chain(underlying, expiry) → Full chain with Greeks
get_futures_chain(symbol) → Available futures contracts
subscribe_market_data(contracts, callback) → Real-time streaming
get_order_book(contract, depth) → Level 2 market depth

# Order Management
place_order(contract, order) → Submit order, return trade object
place_bracket_order(contract, entry, stop, target) → OCO bracket
place_combo_order(legs) → Multi-leg options order
modify_order(trade, new_order) → Modify existing order
cancel_order(trade) → Cancel order
cancel_all_orders() → Cancel everything
get_open_orders() → All pending orders
get_executions() → Today's fills

ontract Definitions:
# Stock
stock = Stock('AAPL', 'SMART', 'USD')

# Option
option = Option('AAPL', '20250221', 185, 'C', 'SMART')
# Symbol, Expiry (YYYYMMDD), Strike, Right (C/P), Exchange

# Future
future = Future('ES', '202503', 'CME')
# Symbol, Expiry (YYYYMM), Exchange

# Futures Option
fut_opt = FuturesOption('ES', '20250221', 5000, 'C', 'CME')

onnection Parameters:
```

```
host = '127.0.0.1'
port = 7497        # Paper trading (7496 for live — NEVER use live without extensive testing)
clientId = 1       # Unique per connection (use different IDs for multiple connections)
timeout = 30       # Connection timeout in seconds
readonly = False   # We need to place orders

CRITICAL IBKR API RULES:
- Maximum 50 simultaneous market data subscriptions
- Maximum 60 historical data requests per 10 minutes
- Maximum 50 orders per second
- Maintain pacing: Don't burst API calls, space them out
- TWS/Gateway must be running at all times during trading
- Paper account simulates realistic fills (not instant)
- Options data returns Greeks (delta, gamma, theta, vega, IV)
```

4.2 IBKR Contract Definitions for This System

```
Contracts We Will Trade:

EQUITIES:
- Dynamic universe of 20-30 liquid stocks (see Universe Selection)
- SPY, QQQ, IWM (ETFs for market direction)
- Sector ETFs: XLK, XLF, XLE, XLV, XLI, XLC, XLY, XLP, XLU, XLRE, XLB

OPTIONS:
- SPX options (European-style, cash-settled, for 0DTE)
- SPY options (American-style, cheaper, also has 0DTE Mon/Wed/Fri)
- QQQ options (daily expirations available)
- Individual stock options on high-IV names:
  TSLA, NVDA, AMD, AMZN, META, GOOG, AAPL, NFLX, COIN, MSTR
- VIX options (for volatility trading)

FUTURES:
- ES (E-mini S&P 500) — $50 per point, trades ~23 hours
- MES (Micro E-mini S&P) — $5 per point (1/10th of ES, good for smaller accounts)
- NQ (E-mini Nasdaq-100) — $20 per point
- MNQ (Micro Nasdaq) — $2 per point
- CL (Crude Oil) — $1,000 per point (very volatile)
- MCL (Micro Crude) — $100 per point
- GC (Gold) — $100 per point
- MGC (Micro Gold) — $10 per point

Contract Creation Helpers:
def make_stock(symbol):
    return Stock(symbol, 'SMART', 'USD')

def make_option(symbol, expiry, strike, right):
    return Option(symbol, expiry, strike, right, 'SMART')

def make_spx_option(expiry, strike, right):
    return Option('SPX', expiry, strike, right, 'CBOE')

def make_future(symbol, expiry, exchange):
    return Future(symbol, expiry, exchange)

def make_micro_future(symbol, expiry):
    exchanges = {'MES': 'CME', 'MNQ': 'CME', 'MCL': 'NYMEX', 'MGC': 'COMEX'}
    return Future(symbol, expiry, exchanges[symbol])
```

---

5. MODULE 2: DATA PIPELINE

5.1 Market Data Collection — All Instruments

5.1.1 Equity Data (via IBKR + yfinance)

```
Real-Time (IBKR Websocket):
- 5-second bars for actively traded symbols
- Real-time quotes (bid/ask/last/volume)
- Market depth (Level 2) for top 5 symbols by signal strength

Historical (IBKR + yfinance):
- IBKR: Intraday data up to 1 year (1-min, 5-min, 15-min, 1-hour bars)
- yfinance: Daily data 5+ years (for ML training and backtesting)
- Merge sources: IBKR for recent intraday, yfinance for long-term daily

Data Fields Per Bar:
timestamp, open, high, low, close, volume, vwap, bar_count,
average_price (from IBKR), trade_count
```

5.1.2 Options Data (via IBKR)

```
Options Chain Data — Pull for each traded underlying:

For Each Option Contract:
- Symbol, expiry, strike, right (call/put)
- Bid, ask, last, mid price
- Volume (today)
- Open interest
- Implied Volatility (IV)
- Greeks: delta, gamma, theta, vega, rho
- Theoretical price (from IBKR's model)

Options Chain Collection Schedule:
- Full chain snapshot: Every 5 minutes during market hours
- 0DTE options: Every 1 minute (they move fast)
- Focus strikes: ATM ± 10 strikes for 0DTE
- Focus expiries: Today (0DTE), this week, next week, monthly

Options Surface Data:
- IV by strike (for skew analysis)
- IV by expiration (for term structure)
- Put/Call ratio by strike
- Put/Call volume ratio
- Max pain calculation (strike with most OI)

Data to Derive:
- IV Rank: Current IV vs. 52-week range → (current - low) / (high - low)
- IV Percentile: % of days in past year with lower IV
- Skew: IV of 25-delta put vs. 25-delta call
- Term Structure: IV by expiration (contango vs. backwardation)
- Put/Call Ratio: Volume and OI based
- Unusual Options Activity: Volume > 5x average OI
```

5.1.3 Futures Data (via IBKR)

```
For Each Futures Contract:
- Real-time bars (1-min, 5-min)
- Bid/ask/last
- Volume
- Open interest
- Fair value vs. cash index (basis)

Futures Specific Data:
- ES Fair Value = SPX + (risk-free rate × days to expiry / 365) × SPX - expected dividends
- Basis = Futures price - Cash index price
- Calendar spread: Front month - Back month
- Roll dates: Track contract roll (quarterly: March, June, Sep, Dec)

Futures Market Hours:
- ES/NQ: Sunday 6 PM ET → Friday 5 PM ET (nearly 24h)
- Only closed: 5 PM → 6 PM ET daily (1 hour)
- This means you can trade futures pre-market and after hours
```

5.1.4 News & Sentiment Data

```
Sources (all free):

1. IBKR News (included with paper account):
   - Real-time news feed through TWS API
   - Covers major news providers
   - Filterable by symbol

2. RSS Feeds (collect every 5 minutes):
   - Reuters Business: http://feeds.reuters.com/reuters/businessNews
   - CNBC: https://www.cnbc.com/id/100003114/device/rss/rss.html
   - MarketWatch: http://feeds.marketwatch.com/marketwatch/topstories
   - Seeking Alpha: https://seekingalpha.com/market_currents.xml
   - Yahoo Finance
   - Bloomberg (RSS excerpts)

3. Reddit API (free tier, every 15 minutes):
   - r/wallstreetbets — retail options sentiment (critical for 0DTE plays)
   - r/options — options-specific discussion
   - r/stocks, r/investing

4. CBOE Data (free, scrape daily):
   - VIX level and term structure
   - Put/Call ratios
   - Total options volume
```

5.1.5 Macro Economic Data (via FRED)

```
Library: fredapi (free API key required)

Key Indicators:
  - FEDFUNDS: Federal Funds Rate
  - DGS10: 10-Year Treasury Yield
  - DGS2: 2-Year Treasury Yield
  - T10Y2Y: 10Y-2Y Spread (yield curve)
  - VIXCLS: VIX Close
  - ICSA: Initial Jobless Claims (weekly)
  - UMCSENT: Consumer Sentiment
  - CPIAUCSL: CPI (monthly)
  - UNRATE: Unemployment (monthly)
  - A191RL1Q225SBEA: GDP Growth (quarterly)
  - BAMLH0A0HYM2: High Yield Credit Spread (risk appetite)
  - DFF: Daily Fed Funds Effective Rate

Collection: Daily at 8:00 AM ET (most update daily or weekly)
Purpose: Macro regime detection + LLM context for market analysis
```

5.2 Data Storage Schema

```
Database: SQLite (file: data/trading.db)
Secondary: Parquet files for large historical datasets


-- Core price data
CREATE TABLE price_bars (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL,
    instrument_type TEXT NOT NULL,  -- 'stock', 'etf', 'future'
    timestamp DATETIME NOT NULL,
    timeframe TEXT NOT NULL,  -- '1min', '5min', '15min', '1hour', '1day'
    open REAL NOT NULL,
    high REAL NOT NULL,
    low REAL NOT NULL,
    close REAL NOT NULL,
    volume INTEGER NOT NULL,
    vwap REAL,
    trade_count INTEGER,
    source TEXT DEFAULT 'ibkr',
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    UNIQUE(symbol, timestamp, timeframe, instrument_type)
);

CREATE INDEX idx_bars_symbol_time ON price_bars(symbol, timestamp, timeframe);


-- Options chain snapshots
CREATE TABLE options_data (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
```

```
    underlying TEXT NOT NULL,
    expiry TEXT NOT NULL,       -- 'YYYYMMDD'
    strike REAL NOT NULL,
    right TEXT NOT NULL,        -- 'C' or 'P'
    timestamp DATETIME NOT NULL,
    bid REAL,
    ask REAL,
    last REAL,
    mid REAL,
    volume INTEGER,
    open_interest INTEGER,
    implied_vol REAL,
    delta REAL,
    gamma REAL,
    theta REAL,
    vega REAL,
    rho REAL,
    theoretical_price REAL,
    days_to_expiry INTEGER,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    UNIQUE(underlying, expiry, strike, right, timestamp)
);

CREATE INDEX idx_options_underlying ON options_data(underlying, expiry, timestamp);

-- IV surface snapshots (aggregated)
CREATE TABLE iv_surface (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    underlying TEXT NOT NULL,
    timestamp DATETIME NOT NULL,
    expiry TEXT NOT NULL,
    atm_iv REAL,
    skew_25d REAL,      -- 25-delta put IV minus 25-delta call IV
    skew_10d REAL,      -- 10-delta skew
    iv_rank REAL,       -- 0-100
    iv_percentile REAL, -- 0-100
    put_call_ratio REAL,
    total_volume INTEGER,
    total_oi INTEGER,
    max_pain_strike REAL,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Futures data
CREATE TABLE futures_bars (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    symbol TEXT NOT NULL,
    contract_month TEXT NOT NULL, -- 'YYYYMM'
    exchange TEXT NOT NULL,
    timestamp DATETIME NOT NULL,
    timeframe TEXT NOT NULL,
    open REAL NOT NULL,
    high REAL NOT NULL,
    low REAL NOT NULL,
    close REAL NOT NULL,
    volume INTEGER NOT NULL,
    open_interest INTEGER,
    basis REAL,  -- futures price - cash index
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    UNIQUE(symbol, contract_month, timestamp, timeframe)
);

-- News and sentiment
CREATE TABLE news (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    headline TEXT NOT NULL,
    summary TEXT,
    source TEXT,
    url TEXT,
    symbols TEXT,           -- JSON array
    published_at DATETIME,
    sentiment_score REAL,   -- -1.0 to 1.0
    sentiment_magnitude REAL,
    catalyst_type TEXT,
```

```
    ai_analysis TEXT,       -- Full JSON from LLM
    options_impact TEXT,    -- JSON: expected IV impact, strategy suggestions
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Macro indicators
CREATE TABLE macro_indicators (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    indicator TEXT NOT NULL,
    date DATE NOT NULL,
    value REAL NOT NULL,
    previous_value REAL,
    UNIQUE(indicator, date)
);

-- Trade log (expanded for options/futures)
CREATE TABLE trades (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    order_id TEXT,
    symbol TEXT NOT NULL,
    instrument_type TEXT NOT NULL,  -- 'stock', 'option', 'future'
    option_expiry TEXT,
    option_strike REAL,
    option_right TEXT,
    future_month TEXT,
    side TEXT NOT NULL,
    quantity REAL NOT NULL,
    price REAL,
    filled_price REAL,
    status TEXT,
    strategy TEXT,
    signal_strength REAL,
    entry_reason TEXT,      -- JSON
    exit_reason TEXT,
    pnl REAL,
    greeks_at_entry TEXT,   -- JSON: delta, gamma, theta, vega at entry
    iv_at_entry REAL,
    iv_at_exit REAL,
    slippage_bps REAL,
    created_at DATETIME DEFAULT CURRENT_TIMESTAMP,
    filled_at DATETIME,
    closed_at DATETIME
);

-- Strategy performance
CREATE TABLE strategy_performance (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    strategy TEXT NOT NULL,
    date DATE NOT NULL,
    trades_taken INTEGER,
    wins INTEGER,
    losses INTEGER,
    total_pnl REAL,
    avg_win REAL,
    avg_loss REAL,
    max_drawdown REAL,
    sharpe_ratio REAL,
    UNIQUE(strategy, date)
);

-- Market regime
CREATE TABLE market_regimes (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    timestamp DATETIME NOT NULL,
    regime TEXT NOT NULL,
    confidence REAL,
    vix_level REAL,
    iv_regime TEXT,  -- 'low_iv', 'normal_iv', 'elevated_iv', 'extreme_iv'
    trend_strength REAL,
    volatility_percentile REAL,
    recommended_options_strategy TEXT
);
```

```
Portfolio Greeks tracking (snapshot every 5 min)
REATE TABLE portfolio_greeks (
 id INTEGER PRIMARY KEY AUTOINCREMENT,
 timestamp DATETIME NOT NULL,
 total_delta REAL,
 total_gamma REAL,
 total_theta REAL,
 total_vega REAL,
 net_liquidation REAL,
 margin_used REAL,
 margin_available REAL,
 buying_power REAL,
 daily_pnl REAL
```

5.3 Data Pipeline Implementation

```
lass: DataPipeline

nitialization:
 1. Connect to IBKR via IBKRClient
 2. Initialize SQLite database (create tables if not exist)
 3. Start background data collection threads:
    - Thread 1: Real-time equity bars via IBKR streaming
    - Thread 2: Options chain snapshots (every 1-5 min)
    - Thread 3: News collection (IBKR + RSS every 5 min)
    - Thread 4: Macro data check (daily at 8 AM)
    - Thread 5: Futures data streaming

Methods:
 start() → Start all data collection streams
 stop() → Gracefully stop all streams, close connections

 # Equities
 get_bars(symbol, timeframe, start, end) → DataFrame
 get_latest_quote(symbol) → {bid, ask, last, volume}
 get_snapshot(symbol) → Full current state
 subscribe_bars(symbols, timeframe, callback)

 # Options
 get_options_chain(underlying, expiry=None) → DataFrame of all options
 get_option_quote(underlying, expiry, strike, right) → Single option data + Greeks
 get_iv_surface(underlying) → IV across strikes and expiries
 get_0dte_chain(underlying) → Today's expiration options only
 get_unusual_options_activity(min_volume_ratio=5) → High volume options

 # Futures
 get_futures_bars(symbol, contract, timeframe, start, end) → DataFrame
 get_futures_quote(symbol, contract) → Current quote + basis
 get_front_month_contract(symbol) → Active front-month contract

 # News
 get_news(symbol=None, since=None) → List of articles
 get_sentiment(symbol, window='1hour') → Aggregated sentiment score

 # Macro
 get_macro_data(indicator) → Latest value + history

Data Validation Rules:
 - Reject bars with volume = 0 (except pre-market)
 - Reject options with bid = 0 and ask = 0
 - Reject options where ask < bid
 - Flag options with spread > 20% of mid price as illiquid
 - Validate Greeks: delta should be between -1 and 1
 - Validate IV: should be between 0.01 and 5.0 (1% to 500%)
 - Timestamps must be timezone-aware (US/Eastern)
 - Deduplicate streaming data
```

5.4 Stock & Options Universe Selection

```
Run daily at 8:00 AM ET (pre-market scan):

EQUITY UNIVERSE (for stock + stock options strategies):
 Criteria:
```

```
  1. Price: $10 - $500
  2. Average daily volume: > 2,000,000 shares
  3. Average daily dollar volume: > $50,000,000
  4. Bid-ask spread: < 0.05% of price
  5. Options available with weekly expirations
  6. Option spreads: ATM option spread < $0.10 or < 5% of option price
  7. ATR(14) > $1.00 (enough movement)
  8. Not in earnings blackout (unless running earnings strategy)
  9. Pre-market relative volume > 1.2x (some unusual interest)
Max Universe: 25 stocks
Min Universe: 10 stocks

0DTE OPTIONS UNIVERSE:
Only certain underlyings have daily (0DTE) expirations:
  - SPX: Monday through Friday
  - SPY: Monday, Wednesday, Friday
  - QQQ: Monday, Wednesday, Friday
  - IWM: Varies
  - TSLA, AAPL, AMZN, NVDA, META, GOOG: Check availability
For 0DTE strategies, focus primarily on SPX and SPY.
SPX advantages: Cash-settled (no assignment risk), European-style, larger notional
SPY advantages: Cheaper per contract, American-style, more liquid at tight strikes

HIGH-IV OPTIONS TARGETS:
Scan for stocks with:
  - IV Rank > 50 (elevated IV relative to own history)
  - High options volume (>10,000 contracts/day)
  - Upcoming catalysts (earnings, FDA, etc.)
These are candidates for volatility selling strategies.

FUTURES UNIVERSE:
Always track:
  - ES or MES (S&P 500 futures)
  - NQ or MNQ (Nasdaq futures)
Optional (if enough compute/data bandwidth):
  - CL or MCL (crude oil — for volatility plays)
  - GC or MGC (gold — macro hedge)

Universe Refresh Schedule:
- Full rebuild: 8:00 AM ET daily
- Quick update: 9:35 AM (after open), 12:00 PM, 2:00 PM
- Add: Stocks that gap > 3% on high volume
- Remove: Stocks with deteriorating liquidity or too-wide options spreads
```

---

6. MODULE 3: FEATURE ENGINEERING

6.1 Equity Technical Indicators

Calculate for each equity across multiple timeframes (1-min, 5-min, 15-min, 1-hour):

```
TREND INDICATORS:
  SMA(10, 20, 50, 200)
  EMA(9, 21, 55)
  MACD(12, 26, 9) — histogram, signal, MACD line
  ADX(14) — trend strength
  Supertrend(10, 3) — trend direction with built-in stop
  Ichimoku Cloud — all 5 lines
  VWAP — intraday anchor
  VWAP Standard Deviation Bands (1σ, 2σ, 3σ)
  Linear Regression Slope(20) — trend steepness

MOMENTUM INDICATORS:
  RSI(14)
  Stochastic RSI(14, 14, 3, 3)
  Williams %R(14)
  CCI(20) — Commodity Channel Index
  ROC(10) — Rate of Change
  MFI(14) — Money Flow Index
  TSI(25, 13) — True Strength Index
  Ultimate Oscillator(7, 14, 28)

VOLATILITY INDICATORS:
  Bollinger Bands(20, 2) — upper, lower, bandwidth, %B
  ATR(14)
  Keltner Channels(20, 2)
  Donchian Channels(20)
  Historical Volatility(20) — annualized
  Bollinger Band Squeeze (BB inside Keltner)
  Garman-Klass Volatility Estimator

VOLUME INDICATORS:
  OBV — On Balance Volume
  VWAP deviation
  Volume Profile — POC, VAH, VAL
  Accumulation/Distribution
  Chaikin Money Flow(21)
  Relative Volume (vs. time-of-day average)
  Volume Rate of Change

SUPPORT/RESISTANCE:
  Pivot Points (Standard, Fibonacci)
  Previous day high/low/close
  Pre-market high/low
  Opening range (first 15 min) high/low
  Fibonacci retracements

PATTERN RECOGNITION:
  Double top/bottom, head & shoulders, triangles,
  flags, inside/outside bars, doji, hammer, engulfing,
  gap detection
```

6.2 Options-Specific Features (NEW)

```
These features are critical for options trading and don't exist in equity-only systems:

PER-UNDERLYING OPTIONS FEATURES:
  atm_iv: At-the-money implied volatility (closest strike to current price)
  iv_rank: IV rank (0-100) vs. 52-week range
  iv_percentile: % of days with lower IV in past year
  iv_change_1d: Change in ATM IV vs. yesterday
  iv_change_5d: 5-day IV change
  hv_iv_spread: Historical volatility minus implied volatility (HV - IV)
    Positive = IV is cheap (buy premium)
    Negative = IV is expensive (sell premium)

  skew_25d: 25-delta put IV - 25-delta call IV
    High skew = fear/hedging demand (bearish signal)
    Low/negative skew = complacency (potential reversal)
  skew_10d: 10-delta skew (extreme wings)
  skew_change_1d: Change in skew

  term_structure_slope: Near-term IV vs. far-term IV
    Contango (far > near): Normal, stable
    Backwardation (near > far): Fear, event-driven, potential opportunity
```

```
put_call_volume_ratio: Total put volume / total call volume
  > 1.0 = bearish sentiment
  < 0.7 = bullish sentiment
  Extreme readings (>1.5 or <0.5) can be contrarian signals

put_call_oi_ratio: Put OI / Call OI
max_pain: Strike price where options buyers lose the most
max_pain_distance: (current_price - max_pain) / current_price

unusual_activity_score: Detect unusual options activity
  = Σ(contracts where volume > 3 × open_interest) weighted by delta
  High score = smart money positioning

gamma_exposure: Net gamma at each strike × open interest
  GEX = Σ(call_gamma × call_oi - put_gamma × put_oi) × 100 × price²
  High positive GEX = market will be "pinned" (dealers hedge → mean reversion)
  Negative GEX = dealers amplify moves (momentum/trend)

vanna_exposure: Sensitivity of delta to changes in IV
  Important for understanding how IV changes will move prices

charm: Rate of delta decay (theta of delta)
  Critical for 0DTE where delta changes rapidly

0DTE-SPECIFIC FEATURES:
dte_0_atm_iv: Current ATM IV for today's expiration
dte_0_volume: Total 0DTE volume
dte_0_pc_ratio: 0DTE put/call ratio
dte_0_gamma_wall: Strike with highest 0DTE gamma exposure
dte_0_expected_move: ATM straddle price as % of underlying
dte_0_time_value_decay_rate: How fast theta is eroding (accelerates near close)
minutes_to_expiry: Exact minutes until 4 PM ET
theta_acceleration: Rate of change of theta (2nd derivative)
```

6.3 Futures-Specific Features

```
basis: Futures price - Cash index price
basis_pct: basis / cash price × 100
basis_z_score: (current basis - mean basis) / std basis
fair_value: Calculated theoretical fair value
premium_discount: Futures price - fair value
calendar_spread: Front month - Next month price
volume_front_vs_back: Front month volume / back month volume
open_interest_change: Daily change in OI (new positions)
cot_positioning: Commitment of Traders data (weekly, from CFTC)
overnight_range: High-low during Asian/European sessions
globex_vwap: VWAP from overnight session
```

6.4 Cross-Asset & Composite Features

```
CROSS-ASSET:
spy_correlation_30d: Rolling 30-day correlation with SPY
sector_relative_strength: Stock return - sector ETF return
market_breadth: NYSE advance/decline ratio
vix_percentile: VIX vs. 1-year distribution
yield_curve_slope: 10Y - 2Y treasury yield
credit_spread: High yield spread (risk appetite)
dollar_index_trend: DXY direction (affects international stocks)
bitcoin_correlation: Rolling BTC-SPY correlation (risk sentiment)

COMPOSITE / DERIVED:
trend_alignment: Agreement across timeframes (-1 to +1)
ma_stack_score: Moving average ordering quality
momentum_composite: Normalized average of RSI, Stoch, MFI, CCI
divergence_score: Price vs. momentum divergences
volatility_regime: ATR percentile in 60-day distribution
squeeze_score: BB/Keltner squeeze tightness
volume_climax: Volume spike with potential reversal
smart_money_flow: Large block trade analysis

# Options-enhanced composites
iv_momentum_divergence: IV falling while price rising (bullish) or vice versa
gamma_regime_score: Net gamma exposure classification
```

```
options_sentiment_composite: Weighted combination of P/C ratio, skew, unusual activity
premium_buyer_vs_seller: Net premium bought vs. sold (from options flow)

TIME FEATURES:
minutes_since_open, minutes_until_close
time_bucket: morning_rush, midday, power_hour
day_of_week, is_monthly_opex, is_fomc_day
is_0dte_expiry_day: True for SPX daily, SPY on M/W/F
days_to_next_earnings: For individual stocks
minutes_to_0dte_expiry: Exact time until today's options expire
```

6.5 Feature Engine Implementation

```
Class: FeatureEngine

Methods:
calculate_equity_features(symbol, bars_df) → DataFrame with all equity features
calculate_options_features(underlying) → Options-specific features
calculate_futures_features(symbol, contract) → Futures features
calculate_cross_asset(symbol) → Cross-asset features
calculate_composite(all_features_df) → Derived/composite features

get_feature_vector(symbol, instrument_type='stock') → Flat numpy array for ML
get_options_feature_vector(underlying) → Options feature array

normalize_features(features_df) → Normalized to [-1, 1] or [0, 1]
get_feature_importance() → Ranking from trained models

Requirements:
- All features normalized for ML consumption
- Handle NaN from insufficient history (expanding window)
- Incremental calculation (update, don't recalculate)
- In-memory cache with TTL
- Options Greeks calculated locally when IBKR data is delayed
  (use Black-Scholes via py_vollib as backup)
```

---

7. MODULE 4: AI/ML BRAIN

7.1 Local LLM via Ollama

```
Setup (already done in Step 4 of installation):
ollama pull llama3.1:8b       # Primary model
ollama pull mistral:7b        # Secondary/ensemble
ollama pull nomic-embed-text  # Embeddings

Ollama API Interface:
Base URL: http://localhost:11434

POST /api/generate — Single prompt completion
POST /api/chat — Multi-turn conversation
POST /api/embeddings — Text embeddings

Parameters for all calls:
  model: "llama3.1:8b"
  temperature: 0.1  # Low for consistent analysis
  num_predict: 2048  # Max tokens
  stream: false  # Wait for full response

Wrapper Class: OllamaClient
  Methods:
    generate(prompt, system_prompt=None) → str
    chat(messages) → str
    embed(text) → list[float]
    analyze_json(prompt, system_prompt) → dict  # Auto-parses JSON response
```

7.1.1 News Analysis Agent

```
For each news article, analyze with LLM:

SYSTEM PROMPT:
"You are a senior quantitative trading analyst specializing in options and
high-volatility instruments. Analyze news for impact on stock prices AND
implied volatility. Respond ONLY in valid JSON with no additional text."
```

```
USER PROMPT:
Analyze this news for trading implications:

Headline: {headline}
Summary: {summary}
Related Symbols: {symbols}
Published: {published_at}

Current State of {primary_symbol}:
Price: ${current_price}
ATM IV: {atm_iv}%
IV Rank: {iv_rank}
IV Percentile: {iv_percentile}
Options P/C Ratio: {pc_ratio}
Recent 0DTE Volume: {dte0_volume}

Market Context:
Regime: {regime}
VIX: {vix}
Sector: {sector}
SPY Change Today: {spy_change}%

Respond in this exact JSON format:
{
  \"sentiment\": <float -1.0 to 1.0>,
  \"magnitude\": <float 0.0 to 1.0>,
  \"time_horizon\": \"<immediate|short_term|medium_term>\",
  \"affected_symbols\": [\"list\"],
  \"catalyst_type\": \"<earnings|macro|sector|company_specific|regulatory|geopolitical|fda|legal>\",
  \"expected_price_move_pct\": <float>,
  \"expected_iv_impact\": \"<iv_crush|iv_spike|iv_unchanged>\",
  \"expected_iv_change_pct\": <float, e.g. 5.0 for 5% increase in IV>,
  \"confidence\": <float 0.0 to 1.0>,
  \"key_factors\": [\"list of 2-3 factors\"],
  \"stock_suggestion\": \"<strong_buy|buy|neutral|sell|strong_sell>\",
  \"options_suggestion\": {
    \"strategy\": \"<buy_calls|buy_puts|sell_puts|sell_calls|straddle|strangle|iron_condor|vertical_spread|none>\",
    \"reasoning\": \"Why this options strategy fits\",
    \"target_expiry\": \"<0dte|weekly|monthly>\",
    \"target_strike_delta\": <float, e.g. 0.30 for 30-delta>
  },
  \"reasoning\": \"1-2 sentence explanation\"
}

Process response:
- Parse JSON, validate all fields
- If confidence > 0.7 and magnitude > 0.5 → Flag as actionable signal
- Store in news table with full ai_analysis
- If expected_iv_impact == 'iv_spike' → Alert volatility strategies
- If catalyst_type == 'earnings' → Check earnings straddle strategy
- Aggregate per symbol: rolling 5min, 1hr, 1day sentiment
```

7.1.2 Market Regime Analysis Agent (Enhanced for Options)

```
Run every 30 minutes during market hours:

SYSTEM PROMPT:
"You are a macro strategist and volatility expert at a quantitative hedge fund.
Classify the current market regime with emphasis on volatility conditions and
optimal options strategies. Respond ONLY in valid JSON."

USER PROMPT:
Classify current market regime:

MARKET DATA:
SPY: ${spy_price}, Today: {spy_change}%, SMA20: ${spy_sma20}, SMA50: ${spy_sma50}
SPY RSI(14): {spy_rsi}, ADX(14): {spy_adx}
Market Breadth (A/D): {breadth}

VOLATILITY DATA:
VIX: {vix}, VIX Change Today: {vix_change}%
VIX 20-day Percentile: {vix_percentile}%
VIX Term Structure: {vix_term_structure} (contango/backwardation)
```

```
SPY ATM IV: {spy_iv}%, IV Rank: {spy_iv_rank}
SPY Skew (25d): {spy_skew}
Total Market Put/Call Ratio: {market_pc_ratio}
0DTE Volume Today: {dte0_volume}

MACRO:
10Y Yield: {yield_10y}%, 2Y Yield: {yield_2y}%
Yield Curve: {yield_curve_slope}bps
Credit Spreads: {credit_spread}bps
DXY: {dollar_index}

GAMMA EXPOSURE:
SPY Net GEX: {spy_gex} (positive = pinning, negative = volatile)
GEX Flip Level: ${gex_flip}

RECENT NEWS:
{recent_news_summary}

Respond in this exact JSON format:

{
\"regime\": \"<bull_trending|bull_volatile|bear_trending|bear_volatile|sideways_calm|sideways_choppy|crisis>\",
\"confidence\": <float 0-1>,
\"trend_direction\": \"<up|down|neutral>\",
\"trend_strength\": <float 0-1>,
\"volatility_level\": \"<low|normal|elevated|extreme>\",
\"iv_regime\": \"<cheap_iv|fair_iv|expensive_iv|extreme_iv>\",
\"risk_appetite\": \"<risk_on|neutral|risk_off|panic>\",
\"gamma_environment\": \"<positive_gamma|neutral|negative_gamma>\",
\"recommended_exposure\": <float 0-1>,
\"recommended_options_strategies\": [\"list of best strategies for this regime\"],
\"avoid_strategies\": [\"strategies to avoid\"],
\"premium_bias\": \"<sell_premium|buy_premium|neutral>\",
\"expected_daily_range\": \"<narrow|normal|wide|extreme>\",
\"key_levels\": {\"spy_support\": X, \"spy_resistance\": Y, \"vix_support\": A, \"vix_resistance\": B},
\"reasoning\": \"Explanation\"
}

REGIME → STRATEGY MAPPING:
bull_trending + low IV → Buy call spreads, sell put spreads, momentum scalping
bull_trending + high IV → Sell put spreads (collect premium + directional)
bear_trending + low IV → Buy put spreads, short momentum
bear_trending + high IV → Sell call spreads, buy protective puts
sideways_calm + low IV → Buy straddles/strangles (expect vol expansion)
sideways_calm + high IV → Sell iron condors (collect premium)
sideways_choppy → Reduce size, wider stops, favor delta-neutral
crisis → Cash, protective puts only, sell covered calls

positive_gamma environment → Market is pinned, mean reversion works
negative_gamma environment → Trends amplify, momentum works
```

7.1.3 Trade Decision Agent (Options-Aware)

```
For each potential trade signal:

SYSTEM PROMPT:
"You are a risk-aware quantitative trader specializing in options and high-volatility
instruments. Evaluate this trade signal considering Greeks exposure, IV conditions,
and portfolio risk. Respond ONLY in valid JSON."

USER PROMPT:
Evaluate this trade signal:

SIGNAL:
Symbol: {symbol}
Instrument: {instrument_type} (stock/option/future)
Strategy: {strategy_name}
Direction: {side}
Signal Strength: {signal_strength}/1.0

ENTRY DETAILS:
Entry Price: ${entry_price}
Stop Loss: ${stop_loss}
Take Profit: ${take_profit}
Risk/Reward: {rr_ratio}
```

```
IF OPTIONS:
Contract: {expiry} {strike}{right}
Current IV: {iv}%
IV Rank: {iv_rank}
Delta: {delta}
Gamma: {gamma}
Theta: {theta}/day
Vega: {vega}
Days to Expiry: {dte}
Spread (bid-ask): ${spread}

MARKET CONTEXT:
Regime: {regime}
IV Regime: {iv_regime}
Gamma Environment: {gamma_env}
VIX: {vix}
Time Until Close: {minutes_to_close} min

PORTFOLIO CONTEXT:
Current Positions: {positions_summary}
Portfolio Delta: {total_delta}
Portfolio Gamma: {total_gamma}
Portfolio Theta: {total_theta}/day
Portfolio Vega: {total_vega}
Daily P&L: ${daily_pnl}
Margin Used: {margin_pct}%
Remaining Risk Budget: ${remaining_risk}
Correlation: {correlation_with_existing}

Respond in this exact JSON format:

{
"decision\": \"<execute|skip|modify>\",
"confidence\": <float 0-1>,
"adjusted_size\": <float 0-1, multiplier>,
"adjusted_stop\": <price or null>,
"adjusted_target\": <price or null>,
"greeks_concerns\": {
  \"delta_impact\": \"<acceptable|concerning|excessive>\",
  \"gamma_risk\": \"<low|moderate|high>\",
  \"theta_decay_concern\": \"<low|moderate|high>\",
  \"vega_exposure\": \"<acceptable|concerning|excessive>\"
},
\"risk_concerns\": [\"list of concerns\"],
\"reasoning\": \"Explanation\"
}
```

7.1.4 End-of-Day Review Agent

```
Run at 4:15 PM ET:

SYSTEM PROMPT:
"You are the chief strategist reviewing today's trading across stocks, options,
and futures. Analyze performance with emphasis on options Greeks management
and volatility plays. Respond ONLY in valid JSON."

USER PROMPT:
"Review today's trading:

Date: {date}
Regime: {regime}, IV Regime: {iv_regime}

TRADES: {trades_json}
(includes: instrument, strategy, entry, exit, greeks at entry/exit, pnl, iv changes)

PERFORMANCE:
Total Trades: {total}, Winners: {winners} ({win_rate}%)
Gross P&L: ${gross_pnl}, Net P&L: ${net_pnl}
Largest Win: ${best} ({best_strategy})
Largest Loss: ${worst} ({worst_strategy})
Profit Factor: {pf}

BY INSTRUMENT TYPE:
Stocks: {stock_pnl}, Options: {options_pnl}, Futures: {futures_pnl}
```

```
GREEKS MANAGEMENT:
- Max Portfolio Delta Today: {max_delta}
- Max Portfolio Gamma Today: {max_gamma}
- Total Theta Collected: ${theta_collected}
- IV Impact on P&L: ${iv_pnl}

OPTIONS SPECIFIC:
- Premium Sold: ${premium_sold}
- Premium Bought: ${premium_bought}
- Net Premium: ${net_premium}
- 0DTE Trades: {dte0_count}, 0DTE P&L: ${dte0_pnl}
- Average IV at Entry vs Exit: {avg_iv_entry}% → {avg_iv_exit}%

MARKET:
- SPY: {spy_change}%, VIX: {vix_change}, Key Events: {events}

Respond in this exact JSON format:

{
  \"overall_grade\": \"<A|B|C|D|F>\",
  \"key_observations\": [\"3-5 observations\"],
  \"greeks_management_grade\": \"<A|B|C|D|F>\",
  \"strategy_adjustments\": [{\"strategy\": \"name\", \"adjustment\": \"what\", \"reason\": \"why\"}],
  \"options_insights\": {
    \"iv_prediction_accuracy\": \"How well did we predict IV moves\",
    \"premium_collection_efficiency\": \"How well did theta work for us\",
    \"0dte_assessment\": \"How did 0DTE trades perform\"
  },
  \"risk_management_notes\": [\"improvements\"],
  \"tomorrow_outlook\": {
    \"expected_regime\": \"<regime>\",
    \"expected_iv_direction\": \"<up|down|stable>\",
    \"recommended_exposure\": <float 0-1>,
    \"recommended_options_bias\": \"<sell_premium|buy_premium|neutral>\",
    \"watch_list\": [\"symbols\"],
    \"key_levels\": {\"spy_support\": X, \"spy_resistance\": Y}
  },
  \"learning_points\": [\"lessons\"]
}
```

7.2 Machine Learning Models

7.2.1 Price Direction Classifier

```
Type: LightGBM (primary) + XGBoost (secondary) ensemble
Purpose: Predict price direction probability over 15/30/60 minutes

Target: Binary (1 if up > 0.2%, 0 if down > 0.2%, exclude flat moves)
Input: Full equity + cross-asset feature vector (~200 features)

Training:
- Min 6 months data, walk-forward validation
- Time-series split (NO future leakage)
- Hyperparameter tuning with Optuna (100 trials)

LightGBM Hyperparameter Ranges:
n_estimators: [100, 1000]
learning_rate: [0.01, 0.3]
max_depth: [3, 10]
num_leaves: [15, 127]
min_child_samples: [5, 100]
subsample: [0.6, 1.0]
colsample_bytree: [0.6, 1.0]
reg_alpha: [0, 10]
reg_lambda: [0, 10]

Schedule: Full retrain weekly, incremental update daily
Output: P(up), P(down), feature importance
```

7.2.2 Volatility Prediction Model

```
Type: LSTM (PyTorch)
Purpose: Predict realized volatility for next 60 min and next day

Architecture:
  Input: 60 time steps × feature vector
  LSTM: 2 layers, 128 hidden, dropout 0.2
  Dense: 128 → 64 → 1

  Train on Apple Silicon GPU via MPS backend:
    device = torch.device("mps" if torch.backends.mps.is_available() else "cpu")

Target: Realized vol (std of returns) over next N periods
Loss: Asymmetric MSE (penalize under-prediction 2x more)
Use: Position sizing, stop loss placement, expected range calculation
```

7.2.3 IV Prediction Model (NEW — Options-Specific)

```
Type: Gradient Boosted Trees (LightGBM)
Purpose: Predict direction of implied volatility change

Input Features:
  - Current IV, IV rank, IV percentile
  - HV-IV spread
  - VIX level and change
  - Skew metrics
  - Time to earnings/events
  - Historical IV patterns (mean reversion tendency)
  - Market regime
  - Recent news sentiment

Target: IV change over next 1 day (up/flat/down classification)

Why This Matters:
  - If IV predicted to rise → Buy premium (straddles, strangles)
  - If IV predicted to fall → Sell premium (iron condors, credit spreads)
  - Combine with direction model:
  - Direction UP + IV UP → Buy calls (best case)
  - Direction UP + IV DOWN → Sell puts (safer premium collection)
  - Direction DOWN + IV UP → Buy puts
  - Direction DOWN + IV DOWN → Sell calls / call spreads
  - Direction NEUTRAL + IV DOWN → Sell straddles/iron condors
```

7.2.4 Sentiment Model (FinBERT)

```
Model: ProsusAI/finbert (local, Hugging Face)
Process: Headline → tokenize → predict [positive, negative, neutral]
Score: P(positive) - P(negative)

Ensemble with Ollama LLM:
  finbert_score: Fast, consistent, good for headlines (0.4 weight)
  llm_score: Deeper context, understands nuance (0.6 weight)
  composite = 0.4 * finbert + 0.6 * llm
```

7.2.5 Regime Classification Model

```
Type: HMM + Random Forest
States: bull_trending, bull_volatile, bear_trending, bear_volatile,
        sideways_calm, sideways_choppy, crisis

Input: SPY returns, volatility, VIX, breadth, yields, credit spreads,
       IV levels, gamma exposure

Combine with LLM regime analysis for final classification
```

7.2.6 Anomaly Detection

```
Isolation Forest + Autoencoder (PyTorch)
When anomaly detected:
  - Reduce position sizes 50%
  - Widen stops
  - Alert LLM for analysis
  - Flag for review
```

7.2.7 Meta-Learner (Ensemble)

```
Stacking Ensemble:
  Layer 1: LightGBM direction, XGBoost direction, LSTM vol, FinBERT sentiment,
           HMM regime, IV predictor, Ollama decision
  Layer 2: Logistic Regression or small NN
  Output: Final trade probability
  Threshold: Only trade if meta-model > 0.65
  Calibration: Platt scaling, track reliability diagram
```

---

8. MODULE 5: TRADING STRATEGIES

EQUITY STRATEGIES (6 Strategies)

Strategy 1: VWAP Mean Reversion

```
Timeframe: 5-min
Concept: Price reverts to VWAP in range-bound markets

LONG ENTRY:
  - Price > 1.5σ below VWAP
  - RSI(14) < 35
  - Relative volume > 1.5x
  - Regime: sideways_calm or sideways_choppy
  - Positive gamma environment (pinning effect helps reversion)
  - Bullish RSI divergence (optional, increases signal strength)

SHORT ENTRY: Mirror above VWAP

Stop: 2σ from VWAP beyond entry or recent swing point
Target: VWAP (T1), 0.5σ opposite side (T2)
Trail: After T1 hit, trail at 1σ from VWAP

Preferred Regimes: sideways_calm, sideways_choppy, positive_gamma
Avoid: Strong trends, negative gamma (amplifies moves away from VWAP)
Hold: 15-45 minutes
```

Strategy 2: Opening Range Breakout (ORB)

```
Timeframe: 15-min (for range), 5-min (for entry)
Concept: First 15-minute range defines the day

SETUP:
  - Opening range = high and low of 9:30-9:45 AM ET
  - Range must be 0.3% to 1.5% of price

LONG BREAKOUT:
  - Price breaks above OR high
  - Volume on breakout > average
  - VWAP rising
  - Enter on pullback to OR high (now support)

Stop: Middle of opening range
Targets: 1x range (T1), 2x range (T2), significant resistance (T3)
Trail: After T1, trail 0.5x range below swing high

Preferred Regimes: Trending markets
Time: 9:45 AM - 11:30 AM ET only
Hold: 30-90 minutes
```

Strategy 3: Momentum Scalp

Timeframe: 5-min
Concept: Ride short-term momentum when indicators align

LONG:
- EMA(9) > EMA(21) on 5-min
- MACD histogram positive and increasing
- RSI(14) between 50 and 70
- Price above VWAP
- Increasing volume over last 3 bars
- ADX(14) > 25
- Sector ETF also trending

Stop: Below EMA(21) or last swing low
Target: ATR(14) × 2
Trail: After +1 ATR, trail at EMA(9)

Preferred Regimes: Trending, negative gamma (amplifies momentum)
Hold: 5-30 minutes, max 1 hour

Strategy 4: Statistical Arbitrage (Pairs)

Concept: Trade spread between correlated/cointegrated stocks

PAIR SELECTION (weekly):
- 60-day correlation > 0.80
- Cointegration test p-value < 0.05
- Calculate spread z-score

ENTRY:
- Z-score > 2.0: Short spread (sell A, buy B)
- Z-score < -2.0: Long spread (buy A, sell B)

Stop: Z-score > 3.0 (relationship breaking)
Target: Z-score reverts to within 0.5 of 0

Preferred Regimes: Sideways (reversion works best)
Hold: 1-5 trading days

Strategy 5: ML Signal Strategy

Concept: Pure ML-driven from ensemble model

ENTRY:
- Meta-model probability > 0.70
- At least 3/5 base models agree
- LLM confirms

Sizing: Proportional to confidence (0.70-0.75: 50%, 0.75-0.85: 75%, 0.85+: 100%)
Stop: predicted_ATR × 1.5
Target: predicted_ATR × 2.5
Hold: Variable (model decides exit too)

Strategy 6: News Catalyst Momentum

Concept: Trade significant news with pullback entry

ENTRY:
- LLM sentiment > 0.7 magnitude > 0.6
- Price moved > 1% in catalyst direction
- Volume > 3x average
- Enter on first pullback (50-61.8% Fib retracement)
- Pullback on decreasing volume

Stop: Below pullback low
Target: 1x initial move (T1), 1.618x Fib extension (T2)
Time: Must enter within 30 min of news
Hold: 30 min - 2 hours

OPTIONS STRATEGIES (6 Strategies — NEW)

Strategy 7: 0DTE Directional Momentum (SPX/SPY)

Timeframe: 1-min and 5-min
Concept: Buy 0DTE calls/puts to capture fast intraday moves with leverage
THIS IS THE HIGHEST RISK / HIGHEST REWARD STRATEGY IN THE SYSTEM

ENTRY CONDITIONS:
- Strong directional signal from equity strategies (signal > 0.75)
- SPX/SPY has clear momentum (ADX > 30, MACD expanding)
- Volume confirmation
- Time: Between 9:45 AM and 2:00 PM ET
(NEVER trade 0DTE after 2 PM unless closing — theta crush is extreme)
- Negative gamma environment (dealers amplify moves)

CONTRACT SELECTION:
- Buy options with 20-30 delta (slightly OTM)
- Strike: First strike where delta is 0.20-0.30
- Spread must be < $0.30 (liquid enough)
- Minimum volume of 1000 contracts at that strike today

POSITION SIZING (CRITICAL — 0DTE can go to $0):
- Maximum 0.5% of portfolio per 0DTE trade
- Maximum 1% total 0DTE exposure at any time
- TREAT THE ENTIRE PREMIUM AS AT RISK

STOP LOSS:
- 50% of premium paid (if bought at $2.00, stop at $1.00)
- This is a HARD stop, no exceptions
- Time stop: Exit if not profitable within 20 minutes

TAKE PROFIT:
- T1: 100% gain (double your money) — sell half
- T2: 200% gain — sell another quarter
- T3: Let rest run with trailing stop at T1 level
- NEVER hold 0DTE past 3:30 PM (theta accelerates exponentially)

TRAILING STOP (after T1):
- Trail 40% below highest value reached

DAILY LIMITS:
- Maximum 3 0DTE directional trades per day
- If first 2 are losers, NO third trade
- Maximum daily 0DTE loss: 1.5% of portfolio → stop 0DTE for the day

WHY THIS WORKS (when it works):
- 0DTE options provide 10-50x leverage on price moves
- A 1% SPX move can = 200-500% return on 0DTE options
- But theta is brutal: options lose ~30-50% of value per hour near close
- The key is TIMING and STRICT RISK MANAGEMENT

Strategy 8: 0DTE Iron Condor (Premium Collection)

```
Timeframe: Daily setup, manage intraday
Concept: Sell OTM puts and calls on 0DTE, profit from theta decay
THIS IS A HIGH-PROBABILITY BUT TAIL-RISK STRATEGY

SETUP:
- Enter between 9:45 AM and 10:30 AM ET
- Sell when VIX > 15 (enough premium to collect)
- Market regime: sideways_calm or sideways_choppy
- Positive gamma environment (market tends to pin)
- No major news catalysts expected today

STRUCTURE:
Sell 1 SPX 0DTE Put at 10-delta (far OTM)
Buy 1 SPX 0DTE Put 5 points lower (wing protection)
Sell 1 SPX 0DTE Call at 10-delta
Buy 1 SPX 0DTE Call 5 points higher (wing protection)

This creates a defined-risk, defined-reward position:
- Max Profit: Premium collected (typically $0.50-$2.00 per spread)
- Max Loss: Width of spread minus premium (typically $3.00-$4.50)
- Breakeven: Short strikes ± premium collected

ENTRY CRITERIA:
- SPX expected move (from ATM straddle) < 1.5%
- Short strikes at least 1.5 standard deviations from current price
- Premium collected > 30% of max loss (decent risk/reward)
- AI regime analysis confirms stable market
- GEX is positive (pinning effect)

MANAGEMENT:
- If SPX moves to 50% of distance to short strike → Close that side
- Take profit at 50% of max premium collected (don't be greedy)
- Time exit: Close all by 3:00 PM ET regardless
- If VIX spikes > 20% intraday → Close immediately
- NEVER hold to expiration (pin risk, assignment risk)

POSITION SIZING:
- Maximum 2% of portfolio at risk per iron condor
- Maximum 2 iron condors open simultaneously
- Maximum daily loss from ICs: 3% of portfolio

WHY THIS WORKS:
- 0DTE options decay rapidly — theta is extreme
- ~70-80% of 0DTE options expire worthless
- Positive GEX days pin the market near current level
- You profit from time decay, collecting small amounts frequently

WHY IT CAN FAIL:
- Unexpected news causes SPX to gap past your strikes
- VIX explosion in negative gamma environment
- That's why the WING PROTECTION and STOP LOSSES are non-negotiable
```

Strategy 9: Gamma Scalping (Delta-Neutral Options)

```
Concept: Buy ATM straddle, then continuously hedge delta to profit from realized volatility

SETUP:
- Buy ATM straddle (call + put at same strike, same expiry)
- Use weekly or 2-week expiration (enough time, reasonable theta)
- Choose underlying with:
  - HV > IV (realized vol exceeds what you're paying in premium)
  - HV-IV spread > 5 percentage points
  - Active enough to move intraday

INITIAL POSITION:
Buy 1 ATM Call + Buy 1 ATM Put = ~0 delta, long gamma, long vega, short theta

GAMMA SCALPING PROCESS (automated):
1. Monitor portfolio delta every 5 minutes
2. When |delta| exceeds threshold (e.g., > 0.30):
   - If delta > +0.30: Sell shares to bring delta back to ~0
   - If delta < -0.30: Buy shares to bring delta back to ~0
3. Each hedge "locks in" the gamma profit from the move
4. Repeat throughout the day

PROFIT SOURCE:
- Long gamma means you make money from price MOVEMENT (not direction)
- Each time you hedge, you lock in a small profit
- As long as realized volatility > implied volatility, you profit net of theta

SIZING:
- Theta cost per day should not exceed 0.5% of portfolio
- Hedge threshold: Re-hedge when delta moves by ±0.3 per option
- Maximum 2 gamma scalping positions simultaneously

EXIT:
- Close when HV/IV ratio falls below 1.0 (no longer favorable)
- Close when remaining time value < cost of continued hedging
- Typically hold 3-7 days
- Close if cumulative theta exceeds cumulative scalping profits

REQUIRED: This strategy requires frequent automated delta hedging.
The system must monitor and hedge every few minutes during market hours.
```

Strategy 10: Volatility Arbitrage (IV vs. HV)

```
Concept: Trade the spread between implied and realized volatility

SELL PREMIUM (IV > HV, IV is expensive):
When:
  - IV Rank > 60
  - IV Percentile > 60
  - HV-IV spread < -10 (IV is 10+ points above HV)
  - No upcoming catalysts (earnings, FDA, etc.)
  - AI regime: sideways preferred

Structures:
  - Short Strangle: Sell OTM put + OTM call (undefined risk — USE WITH CAUTION)
  - Iron Condor: Sell OTM put spread + OTM call spread (defined risk — PREFERRED)
  - Short Straddle: Sell ATM put + call (max theta, high risk)
  - Jade Lizard: Sell put + sell call spread (no upside risk)

Target: 30-45 DTE (theta starts accelerating)
Exit: Close at 50% of max profit OR 21 DTE remaining

BUY PREMIUM (IV < HV, IV is cheap):
When:
  - IV Rank < 20
  - HV-IV spread > 5 (HV exceeds IV)
  - Expecting catalyst or regime change
  - VIX term structure in contango with steep slope

Structures:
  - Long Straddle: Buy ATM call + put
  - Long Strangle: Buy OTM call + put (cheaper but needs bigger move)
  - Calendar Spread: Sell near-term, buy far-term (if term structure is flat)

Target: 30-60 DTE
Exit: Close when IV expands to fair value, or at 50% loss

SIZING:
  - Max 3% of portfolio risk per volatility trade
  - Track portfolio vega: total vega should not exceed 0.5% of portfolio value
```

Strategy 11: Earnings Straddle/Strangle

```
Concept: Trade the known volatility around earnings announcements

PRE-EARNINGS (2-5 days before):
Strategy: Buy straddle/strangle to capture IV expansion into earnings

Entry:
  - 2-5 trading days before earnings
  - Buy weekly expiration that includes earnings date
  - Straddle (ATM) or strangle (5-10% OTM)
  - IV Rank < 70 (want to buy before IV gets too expensive)

Exit (CRITICAL — DO NOT HOLD THROUGH EARNINGS):
  - Sell the day before earnings close or morning of earnings
  - Profit comes from IV expansion, not the earnings move
  - IV typically rises 20-50% into earnings
  - Sell before IV crush destroys your position

POST-EARNINGS (morning after, if there's a big move):
Strategy: Trade the directional move after earnings

Entry:
  - Stock gaps > 5% on earnings
  - Wait for first 15-minute candle to form
  - Buy weekly options in direction of gap if momentum continues
  - OR sell premium if you expect gap fill (put spread against gap up, etc.)

This is similar to the news catalyst strategy but specific to earnings

SIZING:
  - Max 2% of portfolio per earnings trade
  - Max 3 earnings trades in any single week
  - Keep position small: IV crush can destroy 50-80% of option value overnight
```

Strategy 12: The Wheel Strategy (Premium Collection + Stock Accumulation)

oncept: Systematically sell cash-secured puts, then covered calls if assigned
HIS IS THE MOST CONSERVATIVE OPTIONS STRATEGY — suitable for base income

HASE 1: Sell Cash-Secured Puts
- Select stock you'd be happy to own at a lower price
- Sell OTM put (25-35 delta, typically 5-10% below current price)
- Expiration: 30-45 DTE
- Collect premium

Outcomes:
A) Stock stays above strike → Keep premium, sell new put
B) Stock falls below strike → Get assigned, own stock at strike - premium

HASE 2: Sell Covered Calls (if assigned)
- Sell OTM call (25-35 delta, above your cost basis)
- Expiration: 30-45 DTE

Outcomes:
A) Stock stays below strike → Keep premium, sell new call
B) Stock rises above strike → Shares called away at strike + premium profit

TOCK SELECTION:
- Stocks with IV Rank > 30 (enough premium)
- Stocks you actually want to own (quality companies)
- Price < $100 preferred (manageable assignment size, 100 shares)
- Examples: AAPL, AMD, INTC, BAC, F, PLTR, SOFI

IZING:
- Each wheel position = 100 shares (1 contract)
- Max 4 wheel positions running simultaneously
- Max 30% of portfolio in wheel strategies
- Keep enough cash for assignment

ANAGEMENT:
- Roll puts down and out if stock approaches strike and you don't want assignment
- Roll calls up and out if stock approaches strike and you want to keep shares
- Close at 50% profit on premium collected

FUTURES STRATEGY (1 Strategy)

Strategy 13: ES/NQ Micro Futures Scalp

```
Concept: Scalp micro futures for quick moves with leverage

Instruments: MES (Micro S&P, $5/point) or MNQ (Micro Nasdaq, $2/point)
Use micros to keep risk manageable

LONG ENTRY:
- ES/NQ pulls back to VWAP or significant support level
- RSI(14) < 40 on 5-min chart
- Order flow shows buying (uptick ratio > 60%)
- Tick index ($TICK) bouncing from -500 area
- Regime: trending up or sideways with support holding

SHORT ENTRY: Mirror conditions at resistance

ENTRY:
- Limit order at support/resistance level
- Or market order on momentum confirmation

Stop: 5-10 points on MES ($25-$50 per contract)
Target: 10-20 points on MES ($50-$100 per contract)
Trail: After +10 points, trail at 5 points

SIZING:
- Max 4 MES contracts per trade (equivalent to ~$25k notional)
- Max 2% daily loss from futures → stop for the day
- Keep margin usage below 30%

TIMING:
- Best during regular hours: 9:30 AM - 4:00 PM ET
- Also tradeable pre-market (6:00 AM - 9:30 AM) — wider stops, smaller size
- Avoid overnight holds unless swing trading

Hold: 5-30 minutes
```

8.1 Strategy Selection & Dynamic Weighting

```
The system runs all 13 strategies but weights them dynamically:

WEIGHT FACTORS:
1. Market Regime → regime_modifier per strategy
2. IV Regime → options strategies weighted by IV environment
3. Recent Performance → 20-day rolling results
4. Gamma Environment → favor momentum in negative gamma, reversion in positive
5. Time of Day → ORB only early, 0DTE not after 2 PM, etc.
6. Correlation → Avoid taking correlated trades

STRATEGY CATEGORIES:
Equity Directional: Strategies 1-3, 5-6 (can run any time)
Equity Market-Neutral: Strategy 4 (pairs, independent of direction)
Options Directional: Strategy 7 (0DTE momentum, high risk)
Options Premium: Strategies 8, 10, 12 (sell premium, income)
Options Volatility: Strategies 9, 11 (trade vol itself)
Futures: Strategy 13 (fast scalping)

PORTFOLIO ALLOCATION TARGETS:
Equity strategies: 30-50% of capital
Options premium selling: 20-30% of capital
Options directional/vol: 10-20% of capital
Futures: 10-15% of capital
Cash reserve: 10-20% always (for opportunities and margin)

REGIME-BASED ALLOCATION:
Bull Trending + Low IV:
  → Heavy equity momentum + options directional
  → Light premium selling (not enough premium)

Sideways + High IV:
  → Heavy premium selling (iron condors, wheel)
  → Light equity direction
  → Some gamma scalping

Crisis:
```

→ Minimal exposure
→ Only protective puts and cash
→ Reduce to 20% of normal sizing

---

9. MODULE 6: OPTIONS INTELLIGENCE ENGINE (NEW)

This is a dedicated module for options-specific calculations and analysis.

lass: OptionsIntelligence

ORE CALCULATIONS:

Greeks Calculation (when IBKR data is unavailable):
- Black-Scholes for European options (SPX)
- Binomial model for American options (stocks, SPY)
- Library: py_vollib or mibian

def calculate_greeks(S, K, T, r, sigma, option_type):
\"\"\"
S = underlying price
K = strike price
T = time to expiry (years)
r = risk-free rate
sigma = implied volatility
option_type = 'call' or 'put'
Returns: delta, gamma, theta, vega, rho
\"\"\"

IV Surface Construction:
- Build full IV surface (strike × expiry matrix)
- Interpolate for missing strikes
- Calculate ATM IV for each expiry
- Calculate constant-maturity IV (30-day, 60-day, 90-day)

Skew Analysis:
- 25-delta put-call skew
- 10-delta skew (tail risk indicator)
- Skew term structure (how skew changes with expiry)
- Risk reversal prices

Term Structure:
- Plot IV by expiration
- Classify: contango (normal) vs. backwardation (stressed)
- Calculate slope and rate of change

Gamma Exposure (GEX):
- GEX = Σ over all strikes:
(call_gamma × call_OI × 100 - put_gamma × put_OI × 100) × price × price / 1e9
- Calculate for SPY/SPX to determine market pinning behavior
- Positive GEX → market pins → mean reversion
- Negative GEX → moves amplify → momentum
- GEX flip level: price where GEX changes sign

Max Pain:
- For each strike, calculate total loss for all option holders
- Max pain = strike where total losses are maximized
- Market often gravitates toward max pain near expiration

Expected Move:
- From ATM straddle: expected_move = straddle_price / underlying_price
- From IV: expected_move_1d = IV × sqrt(1/252)
- Compare expected move to actual move for edge detection

Unusual Options Activity Detection:
- Flag when: volume > 3 × open interest
- Flag when: large block trade (>100 contracts at single price)
- Flag when: options volume > 3x 20-day average
- Classify: bullish (call buying, put selling) vs. bearish
- LLM analysis of unusual flow for context

Position P&L Modeling:
- For any options position, calculate:
- P&L at current price
- P&L at target price
- P&L at stop price
- P&L across range of prices (payoff diagram)
- Greeks exposure at various prices
- Time decay impact (theta per day)
- IV sensitivity (vega × expected IV change)

Optimal Strike Selection:

```
  - Given strategy and direction, find optimal strike:
    - For buying: Best risk/reward ratio considering delta, spread, volume
    - For selling: Best premium collection considering probability of profit
    - Kelly-optimal delta for directional bets

Roll Decision Engine:
  - When to roll options positions:
    - Time-based: Roll when < 7 DTE for sold options (gamma risk)
    - Profit-based: Roll when > 50% of max profit captured
    - Threat-based: Roll when short strike is tested (price approaches)
    - Suggest roll target: Same strike further out, or different strike

Methods:
get_greeks(underlying, expiry, strike, right) → Greeks dict
build_iv_surface(underlying) → 2D DataFrame
get_skew(underlying, expiry) → Skew metrics
get_term_structure(underlying) → Term structure data
calculate_gex(underlying) → GEX value + flip level
calculate_max_pain(underlying, expiry) → Strike price
get_expected_move(underlying, expiry) → Percentage
detect_unusual_activity(underlying) → List of unusual trades
model_position_pnl(position, price_range) → PnL array
select_optimal_strike(underlying, strategy, direction) → Strike
should_roll(position) → RollDecision
```

---

10. MODULE 7: RISK MANAGEMENT ENGINE (OPTIONS-AWARE)

10.1 Position Sizing

```
EQUITIES:
  Same as before:
  - Max single position: 5% of portfolio
  - Max per-trade risk: 1% of portfolio
  - Half-Kelly sizing with volatility adjustment

OPTIONS — LONG (buying premium):
  - Max premium at risk per trade: 0.5% of portfolio for 0DTE, 1% for weekly+
  - NEVER risk more than the premium paid (options can go to $0)
  - Size by premium, not by notional:
  If buying 10 calls at $2.00 = $2,000 at risk
  This should be < 1% of $100,000 portfolio ✓

OPTIONS — SHORT (selling premium):
  - Max loss per defined-risk trade: 2% of portfolio
  - For undefined risk (naked): Max notional = 5% of portfolio
  - STRONGLY PREFER defined-risk strategies (spreads, iron condors)
  - Always know max loss BEFORE entering

FUTURES:
  - Max per-trade risk: 1% of portfolio
  - Position size = (risk budget) / (stop distance × point value)
  - Example: $1,000 risk / (10 points × $5/point MES) = 20 contracts max
  - Keep margin usage below 30% of account

OPTIONS GREEKS PORTFOLIO LIMITS:
  - |Total Portfolio Deltal < 30% of net liquidation / 100
  (if portfolio = $100k, |deltal < 300, roughly equivalent to 300 shares SPY)
  - |Total Portfolio Gammal < 1% of net liq / 100
  - Total Portfolio Theta: Should be collecting theta, not paying
  Max negative theta: -0.1% of portfolio per day ($100/day on $100k)
  - |Total Portfolio Vegal: < 0.5% of net liq per 1% IV move
```

10.2 Stop Loss Rules (All Instruments)

```
EQUITIES: (same as before)
  - Every trade has a stop, ATR-based, trailing after profit

OPTIONS — LONG:
  - Stop at 50% of premium paid (0DTE)
  - Stop at 40% of premium paid (weekly)
  - Stop at 30% of premium paid (monthly)
  - Time stop: If not profitable in 1/3 of DTE, close
  - HARD RULE: Never hold 0DTE past 3:30 PM ET
```

- HARD RULE: Never hold weekly options through last day unless intentional

OPTIONS — SHORT (credit spreads, iron condors):
- Stop when loss = 2x premium collected
(if you collected $1.00, close at $3.00)
- Stop when short strike delta reaches 0.40 (getting tested)
- Stop when underlying reaches 50% of distance to short strike
- Take profit at 50% of premium collected (don't be greedy)
- Close all positions by 3:00 PM on expiration day

FUTURES:
- Stop in points: 5-15 points on MES depending on volatility
- Always use native stop orders (not mental stops)
- Trail after target 1 hit

10.3 Portfolio-Level Risk Controls

DAILY LIMITS:
- Max daily loss (all instruments): 3% of portfolio → STOP ALL TRADING
- Max daily loss per instrument class:
Equities: 1.5%
Options: 2%
Futures: 1.5%
- Max consecutive losses: 5 → Reduce all sizes 50% for 1 hour
- Max open positions: 8 total across all instruments
- Max correlated positions: 3 in same direction/sector

WEEKLY LIMITS:
- Max weekly loss: 5% → Reduce all sizes 50% next week
- Max weekly drawdown from peak: 7% → Paper trade only until reviewed

MARGIN MONITORING:
- Alert at 50% margin usage
- Reduce positions at 60% margin usage
- Close positions at 70% margin usage
- IBKR will liquidate at ~90% — NEVER get close to this
- Check margin every 5 minutes

GREEKS LIMITS (portfolio-level):
- Max |delta|: 300 (roughly 300 shares SPY equivalent)
- Max |gamma|: Monitor but focus on negative gamma risk
- Max negative theta: -$100/day per $100k
- Max |vega|: $500 per 1% IV change per $100k
- If any limit exceeded: Close or hedge the most offsetting position

CORRELATION:
- Real-time correlation of all positions
- Max portfolio beta: 1.5 (vs. SPY)
- If all options are on same underlying → single point of failure
- Diversify across at least 3 different underlyings

PRE-TRADE CHECKLIST (automated):
□ Position size within limits?
□ Stop loss defined?
□ Daily risk budget available?
□ Margin sufficient?
□ Portfolio Greeks within limits?
□ Not too correlated with existing?
□ Not in restricted time window?
□ Liquidity adequate (bid-ask spread)?
□ LLM confirmation received?
□ For options: defined risk or adequate hedge?

10.4 Circuit Breakers

LEVEL 1 — CAUTION (Yellow):
Trigger: -1.5% daily OR VIX spike > 10% intraday OR margin > 50%
Action:
- New position sizes reduced 50%
- Widen stops 25%
- Only take signals > 0.8 strength
- Close any undefined-risk options positions
- Alert + log

```
LEVEL 2 — WARNING (Orange):
Trigger: -2.5% daily OR VIX > 30 OR margin > 60% OR market halt
Action:
  - Stop all new positions
  - Tighten trailing stops on winners
  - Close any 0DTE positions
  - Alert + log

LEVEL 3 — EMERGENCY (Red):
Trigger: -3% daily OR 3+ consecutive losses in < 30 min OR margin > 70%
Action:
  - Close ALL positions at market
  - Cancel ALL open orders
  - Stop trading for the rest of the day
  - Run LLM post-mortem
  - Alert + log

SPECIAL — OPTIONS CIRCUIT BREAKER:
Trigger: Any single options position loses > 80% of premium (long) or
         reaches 3x premium (short credit position)
Action: Close that position immediately, review strategy
```

---

11. MODULE 8: EXECUTION ENGINE (IBKR-SPECIFIC)

11.1 Order Types via IBKR

```
Using ib_insync:

EQUITY ORDERS:
  MarketOrder(side, quantity)
  LimitOrder(side, quantity, limitPrice)
  StopOrder(side, quantity, stopPrice)
  StopLimitOrder(side, quantity, stopPrice, limitPrice)
  TrailingStopOrder(side, quantity, trailingPercent)

BRACKET ORDERS (for equities and futures):
  Use ib.bracketOrder(action, quantity, limitPrice, takeProfitPrice, stopLossPrice)
  This creates parent + 2 child orders (OCA group)
  ALWAYS use brackets for directional equity/futures trades

OPTIONS ORDERS:
  Single leg: Same order types as above

  Multi-leg (spreads, condors):
    contract = Contract()
    contract.symbol = 'SPX'
    contract.secType = 'BAG'  # Combination/combo order
    contract.exchange = 'SMART'
    contract.currency = 'USD'
    contract.comboLegs = [
        ComboLeg(conId=leg1_conid, ratio=1, action='SELL'),
        ComboLeg(conId=leg2_conid, ratio=1, action='BUY'),
        ComboLeg(conId=leg3_conid, ratio=1, action='SELL'),
        ComboLeg(conId=leg4_conid, ratio=1, action='BUY'),
    ]
    # Submit as LimitOrder with net credit/debit price

  This sends the entire iron condor / spread as a single order
  IBKR handles all legs atomically (no leg risk)

FUTURES ORDERS:
  Same as equity orders but with Future() contract
  IBKR handles margin automatically

ORDER MANAGEMENT:
  - Check order status every 10 seconds via ib.openOrders()
  - Cancel unfilled limits after 2 minutes
  - Handle partial fills: Adjust child orders accordingly
  - Log every order event: submitted, filled, partial, cancelled, rejected, error
  - Handle IBKR error codes gracefully:
    Code 201: Order rejected (insufficient margin)
    Code 202: Order cancelled
```

- Code 110: Invalid price
- Code 201: Duplicate order

11.2 Execution Optimization

```
EQUITIES:
- Small orders (<500 shares): Market order for urgency, limit for patience
- Large orders (>500 shares): TWAP over 2 minutes
- Always use SMART routing (IBKR's best execution router)

OPTIONS:
- ALWAYS use limit orders (never market orders on options)
- Options spreads can be very wide; market orders = terrible fills
- Start at mid price, improve by $0.01 every 15 seconds
- For spreads: Price as net credit/debit, start at mid of natural
- If not filled in 2 minutes, evaluate: improve price or cancel
- For 0DTE: More aggressive pricing (closer to natural) due to time urgency

FUTURES:
- Market orders for scalps (speed matters)
- Limit orders for anticipated levels

SLIPPAGE TRACKING:
Track for every trade:
- expected_fill = mid_price at signal time
- actual_fill = reported fill price
- slippage_bps = |actual - expected| / expected × 10000
Alert if average slippage > 5bps for equities, >10bps for options
```

---

12. MODULE 9: MONITORING & ADAPTIVE LAYER

12.1 Dashboard (Dash/Plotly)

```
Page 1: Live Trading
- Positions table with real-time P&L (color-coded)
- Open orders
- Daily equity curve
- Active signals queue
- Regime indicator + IV regime
- Circuit breaker status lights
- Portfolio Greeks display (delta, gamma, theta, vega)
- Margin utilization bar

Page 2: Options View
- Options positions with Greeks per position
- Portfolio Greeks pie chart
- IV surface heatmap for top 3 underlyings
- Skew chart
- 0DTE status and theta countdown
- Unusual options activity feed

Page 3: Strategy Performance
- Per-strategy equity curves
- Win rate / profit factor / Sharpe by strategy
- By instrument type breakdown
- Regime-strategy heatmap
- Rolling 20-day performance

Page 4: AI Insights
- Latest LLM analysis
- News sentiment feed
- Model confidence gauges
- Feature importance chart
- Anomaly alerts

Page 5: Risk Dashboard
- Position heat map
- Correlation matrix
- Risk budget utilization
- Max drawdown chart
- VaR estimate
- Greeks exposure over time
- Margin history
```

12.2 Adaptive Learning Loop

```
DAILY:
- LLM end-of-day review
- Update strategy weights
- Recalculate feature importance
- Update regime classification
- Log all metrics

WEEKLY:
- Retrain ML models (walk-forward)
- Backtest current vs. last week's parameters
- Update stock/options universe
- Recalculate pair cointegration
- Update IV prediction model
- LLM weekly performance report

MONTHLY:
- Full hyperparameter optimization (Optuna)
- Architecture review (are all strategies viable?)
- Feature engineering review
- LLM prompt optimization
- Options strategy effectiveness review
- Greeks management assessment
```

---

13. MODULE 10: BACKTESTING FRAMEWORK

```
REQUIREMENTS:
- Walk-forward validation (mandatory)
- Time-series split, no future leakage
- Realistic assumptions:
  Equity commissions: $0 (IBKR Lite)
  Options commissions: $0.65 per contract (IBKR standard) — include this!
  Futures commissions: $0.85 per contract (include this)
  Equity slippage: 0.02% per trade
  Options slippage: $0.02-0.05 per contract (spread dependent)
  Options fill rate: Only 60% of limit orders fill
  Assignment risk modeling for American-style options

- Test periods:
  In-sample: 2020-2023 (COVID crash, recovery, rate hikes)
  Out-of-sample: 2024-present
  Stress: March 2020 crash, 2022 bear, Oct 2023 vol spike

- Metrics targets:
  Sharpe > 1.5
  Sortino > 2.0
  Max Drawdown < 20% (higher than equity-only because of options leverage)
  Win Rate > 55%
  Profit Factor > 1.5
  Options: Track Greeks P&L attribution separately

- Overfitting prevention:
  Walk-forward validation (mandatory)
  Out-of-sample test (mandatory)
  Min 500 trades for statistical significance
  If in-sample Sharpe > 2x out-of-sample → likely overfit
  Test across multiple regimes
```

---

14. MODULE 11: SYSTEM CONFIGURATION

14.1 Project Structure

```
ai-trading-system/
├── config/
│   ├── settings.yaml
│   ├── .env
│   └── strategies.yaml
├── src/
│   ├── __init__.py
│   ├── main.py                  # Entry point / orchestrator
```

```
├── data/
│   ├── pipeline.py            # DataPipeline
│   ├── ibkr_client.py         # IBKR connection wrapper
│   ├── news_collector.py
│   ├── macro_data.py
│   └── universe.py
├── features/
│   ├── engine.py              # FeatureEngine
│   ├── technical.py
│   ├── options_features.py    # Options-specific features
│   ├── futures_features.py
│   ├── derived.py
│   └── cross_asset.py
├── ai/
│   ├── llm_client.py          # Ollama wrapper
│   ├── news_agent.py
│   ├── regime_agent.py
│   ├── trade_agent.py
│   ├── review_agent.py
│   └── prompt_templates.py
├── ml/
│   ├── direction_model.py     # LightGBM
│   ├── volatility_model.py    # LSTM
│   ├── iv_model.py            # IV predictor (NEW)
│   ├── sentiment_model.py     # FinBERT
│   ├── regime_model.py        # HMM + RF
│   ├── anomaly_model.py
│   ├── ensemble.py            # Meta-learner
│   └── training.py
├── options/                   # NEW MODULE
│   ├── intelligence.py        # OptionsIntelligence class
│   ├── greeks.py              # Greeks calculations
│   ├── iv_surface.py          # IV surface builder
│   ├── gex.py                 # Gamma exposure
│   ├── unusual_activity.py    # Unusual options activity detector
│   └── pricing.py             # Black-Scholes, binomial
├── strategies/
│   ├── base.py
│   ├── vwap_reversion.py      # Strategy 1
│   ├── orb.py                 # Strategy 2
│   ├── momentum_scalp.py      # Strategy 3
│   ├── pairs.py               # Strategy 4
│   ├── ml_signal.py           # Strategy 5
│   ├── news_catalyst.py       # Strategy 6
│   ├── dte0_momentum.py       # Strategy 7 (NEW)
│   ├── dte0_iron_condor.py    # Strategy 8 (NEW)
│   ├── gamma_scalp.py         # Strategy 9 (NEW)
│   ├── vol_arb.py             # Strategy 10 (NEW)
│   ├── earnings_straddle.py   # Strategy 11 (NEW)
│   ├── wheel.py               # Strategy 12 (NEW)
│   ├── futures_scalp.py       # Strategy 13 (NEW)
│   └── selector.py
├── risk/
│   ├── position_sizer.py
│   ├── stop_manager.py
│   ├── portfolio_risk.py
│   ├── greeks_manager.py      # Portfolio Greeks limits (NEW)
│   ├── margin_monitor.py      # IBKR margin tracking (NEW)
│   └── circuit_breaker.py
├── execution/
│   ├── order_manager.py       # IBKR order execution
│   ├── options_orders.py      # Multi-leg options orders (NEW)
│   ├── smart_router.py
│   └── position_tracker.py
├── monitoring/
│   ├── dashboard.py
│   ├── metrics.py
│   ├── logger.py
│   └── alerts.py
└── utils/
    ├── database.py
    ├── time_utils.py
    └── validators.py
```

```
├── models/
├── data/
│   ├── trading.db
│   └── parquet/
├── logs/
├── notebooks/
├── tests/
├── requirements.txt
└── README.md
```

14.2 Configuration (settings.yaml)

```yaml
```

```yaml
trading:
  mode: paper
  broker: ibkr
  timezone: US/Eastern
  market_open: "09:30"
  market_close: "16:00"
  trading_start: "09:35"
  trading_end: "15:55"
  close_all_eod: true
  instruments:
    equities: true
    options: true
    futures: true

ibkr:
  host: "127.0.0.1"
  port: 7497  # Paper trading
  client_id: 1
  timeout: 30

portfolio:
  initial_capital: 100000
  max_position_pct: 0.05
  max_risk_per_trade_pct: 0.01
  max_options_risk_per_trade_pct: 0.01
  max_0dte_risk_per_trade_pct: 0.005
  max_open_positions: 8
  max_daily_loss_pct: 0.03
  max_weekly_loss_pct: 0.05
  target_annual_volatility: 0.20
  max_margin_usage_pct: 0.50

  # Portfolio Greeks limits
  max_abs_delta: 300
  max_negative_theta_daily: -100  # dollars
  max_abs_vega: 500

  # Allocation targets
  equity_allocation: 0.40
  options_allocation: 0.30
  futures_allocation: 0.15
  cash_reserve: 0.15

data:
  history_days: 730
  bar_timeframes: ['1min', '5min', '15min', '1hour', '1day']
  max_universe_size: 25
  options_chain_refresh_sec: 300  # 5 min
  dte0_chain_refresh_sec: 60     # 1 min for 0DTE

ai:
  ollama_base_url: "http://localhost:11434"
  primary_model: "llama3.1:8b"
  secondary_model: "mistral:7b"
  embedding_model: "nomic-embed-text"
  temperature: 0.1
  regime_analysis_interval_min: 30

ml:
  direction_model: lightgbm
  retrain_schedule: weekly
  meta_model_threshold: 0.65

risk:
  circuit_breaker_yellow: -0.015
  circuit_breaker_orange: -0.025
  circuit_breaker_red: -0.03
  max_consecutive_losses: 5
  max_correlation: 0.7
  max_sector_exposure: 3
  margin_alert_pct: 0.50
  margin_reduce_pct: 0.60
  margin_close_pct: 0.70
```

```yaml
options:
  prefer_defined_risk: true
  max_0dte_trades_per_day: 3
  max_0dte_daily_loss_pct: 0.015
  iron_condor_take_profit_pct: 0.50
  wheel_max_positions: 4
  gamma_scalp_hedge_threshold: 0.30

monitoring:
  dashboard_enabled: true
  dashboard_port: 8050
  log_level: INFO
```

14.3 Requirements (requirements.txt)

```
# Core
python-dotenv>=1.0
pyyaml>=6.0
apscheduler>=3.10

# Interactive Brokers
ib_insync>=0.9.86

# Data
yfinance>=0.2
fredapi>=0.5
pandas>=2.1
numpy>=1.25
pyarrow>=14.0
feedparser>=6.0
praw>=7.7
requests>=2.31
aiohttp>=3.9

# Technical Analysis
TA-Lib>=0.4.28
pandas-ta>=0.3

# Machine Learning
scikit-learn>=1.3
lightgbm>=4.1
xgboost>=2.0
optuna>=3.4
torch>=2.1
hmmlearn>=0.3

# NLP / Sentiment
transformers>=4.35
tokenizers>=0.14

# Options
py_vollib>=1.0
mibian>=0.1.3
scipy>=1.11

# Visualization
plotly>=5.18
dash>=2.14

# Backtesting
vectorbt>=0.26

# Utilities
tqdm>=4.66
structlog>=23.2
```

---

15. MODULE 12: SAFETY & OPERATIONAL RULES

```
NON-NEGOTIABLE RULES:

1. PAPER TRADE MINIMUM 3 MONTHS before considering real money
   - Positive Sharpe > 1.0
```

- Survived at least one vol spike / drawdown
- All circuit breakers tested
- Options strategies validated with real fills and assignment scenarios

2. NEVER store credentials in code. Use .env + .gitignore.

3. ALL day trading positions closed by EOD:
- Equities: By 3:55 PM ET
- 0DTE Options: By 3:30 PM ET (earlier!)
- Futures scalps: By close (unless intentional swing)
- Exception: Wheel positions and multi-week options positions

4. OPTIONS-SPECIFIC SAFETY:
- ALWAYS prefer defined-risk (spreads) over undefined risk (naked)
- NEVER sell naked calls on individual stocks
- ALWAYS know max loss before entering any trade
- 0DTE maximum risk per trade: 0.5% of portfolio
- NEVER hold 0DTE options into the last 30 minutes unless closing
- Check assignment risk daily for American-style short options
- Monitor margin continuously

5. FUTURES SAFETY:
- Use MICRO contracts (MES, MNQ, MCL) not full-size
- Keep margin below 30%
- Always have stops in place

6. SYSTEM MUST HANDLE GRACEFULLY:
- IBKR API disconnection → Reconnect, close positions if prolonged
- Ollama crash → Skip LLM confirmation, use ML-only signals (reduced size)
- Network outage → Close all positions via emergency procedure
- Database lock → Queue writes, never lose trade data
- Invalid data → Reject and log, don't trade on bad data
- Market halt → Pause all strategies, wait for resume
- Options assignment → Handle automatically, adjust portfolio

7. LOGGING:
- Every signal, order, fill, LLM response, model prediction, risk check
- Structured JSON logs for analysis
- Separate log files: trading.log, risk.log, ai.log, errors.log

8. BEFORE ANY CHANGE:
- Backtest first
- Paper trade 1 week minimum
- Compare before/after metrics

9. RESOURCE MANAGEMENT:
- Monitor Ollama RAM usage (kill/restart if memory leak)
- IBKR max 50 simultaneous market data subscriptions
- Pace API calls (max 60 historical requests per 10 min)
- SQLite WAL mode for concurrent reads/writes

---

16. IMPLEMENTATION PRIORITY ORDER

Build in this exact order. Each step should be independently testable.

PHASE 1: FOUNDATION (Week 1-2)
Step 1: Install everything (Section 3.2)
Step 2: Build IBKRClient — connect, get account info, get quotes
Step 3: Build DataPipeline — stream equity bars, store in SQLite
Step 4: Build FeatureEngine — calculate technical indicators
Step 5: Verify: Can you pull real-time SPY data and compute RSI? ✓

PHASE 2: FIRST TRADE (Week 3-4)
Step 6: Build ExecutionEngine — place simple equity bracket orders
Step 7: Build basic RiskManager — position sizing, daily limits
Step 8: Implement Strategy 3 (Momentum Scalp) — simplest strategy
Step 9: Build StopManager — initial stops, trailing stops
Step 10: Paper trade Strategy 3 for 1 week manually verify trades ✓

PHASE 3: INTELLIGENCE (Week 5-7)
Step 11: Build OllamaClient — connect to local LLM
Step 12: Implement News Agent — analyze news with LLM
Step 13: Implement Regime Agent — classify market regime
Step 14: Implement Trade Decision Agent — LLM confirmation
Step 15: Build FinBERT sentiment model
Step 16: Integrate AI into trading loop ✓

PHASE 4: ML MODELS (Week 8-10)
Step 17: Build direction classifier (LightGBM)
Step 18: Build volatility predictor (LSTM)
Step 19: Build regime classifier (HMM)
Step 20: Build meta-learner ensemble
Step 21: Implement Strategy 5 (ML Signal) ✓

PHASE 5: MORE STRATEGIES (Week 11-13)
Step 22: Implement Strategy 1 (VWAP Reversion)
Step 23: Implement Strategy 2 (ORB)
Step 24: Implement Strategy 4 (Pairs)
Step 25: Implement Strategy 6 (News Catalyst)
Step 26: Build strategy selector/weight system ✓

PHASE 6: OPTIONS (Week 14-18)
Step 27: Build OptionsIntelligence — Greeks, IV surface, GEX
Step 28: Build options order execution (multi-leg via IBKR)
Step 29: Build IV prediction model
Step 30: Implement Strategy 12 (Wheel) — safest options strategy, start here
Step 31: Implement Strategy 10 (Vol Arb — iron condors)
Step 32: Implement Strategy 8 (0DTE Iron Condor)
Step 33: Implement Strategy 7 (0DTE Momentum) — most dangerous, add last
Step 34: Implement Strategy 9 (Gamma Scalp)
Step 35: Implement Strategy 11 (Earnings Straddle)
Step 36: Build Greeks portfolio manager ✓

PHASE 7: FUTURES (Week 19-20)
Step 37: Add futures data to DataPipeline
Step 38: Build futures-specific features
Step 39: Implement Strategy 13 (MES/MNQ Scalp)
Step 40: Build margin monitor ✓

PHASE 8: INFRASTRUCTURE (Week 21-22)
Step 41: Build monitoring dashboard (Dash)
Step 42: Build adaptive learning loop
Step 43: Build End-of-Day review agent
Step 44: Build comprehensive backtesting framework
Step 45: Backtest all strategies ✓

PHASE 9: OPTIMIZATION (Week 23-24)
Step 46: Hyperparameter optimization (Optuna)
Step 47: Feature importance analysis and pruning
Step 48: LLM prompt optimization
Step 49: Performance attribution analysis ✓

PHASE 10: EXTENDED PAPER TRADING (Month 7-9+)
Step 50: Run full system in paper trading for 3+ months
Step 51: Monitor, adjust, learn
Step 52: Only after consistent results: Consider real money with MINIMAL size

COST SUMMARY (Final)

Item	Cost
IBKR Paper Account	Free
IBKR Market Data (delayed)	Free
IBKR Real-Time Data (optional)	$0 - $15/month (waived with commissions)
Ollama + Models	Free (local)
FinBERT + Hugging Face	Free (local)
Python + All Libraries	Free
yfinance / FRED	Free
SQLite	Free
Electricity	~$5-10/month
Total	**$0 - $15/month**

For true $0 cost: Use delayed data (15 min) for paper trading. This is fine for testing strategies. Switch to real-time data ($10-15/mo) when you're ready for serious paper trading evaluation.

FINAL NOTE

This system is designed to be built incrementally. Do NOT try to build everything at once. Follow the Phase 1-10 order. Each phase adds capability while the previous phases ensure the foundation is solid.

The most important modules are: **Risk Management** and **Execution**. A system with average signals but excellent risk management will survive. A system with great signals but poor risk management will blow up.

Options and 0DTE trading are EXTREMELY high risk. The leverage cuts both ways. Paper trade extensively before even considering real capital.

This entire blueprint can be handed to an AI coding assistant section by section to implement. Each module is self-contained with clear interfaces, data schemas, and specifications.